

COSIMA Cookbook: a computational learning module for analysing ocean and sea-ice model output

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Summary

The COSIMA Cookbook is an open computational learning module for analysing ocean and sea-ice model output in Jupyter notebooks. It has been developed by the Consortium for Ocean-Sea Ice Modelling in Australia (COSIMA) as a community resource for researchers, students, and practitioners working with large gridded datasets, especially output from the ACCESS ocean-sea ice model configuration (Kiss et al., 2020). The repository combines introductory tutorials with worked analysis examples, all exposed through a browsable Sphinx documentation site and backed by a citable archived release (COSIMA contributors, 2024).

The instructional structure is central to the project. “Cooking Lessons 101” introduces generic skills such as loading model output, working with labelled arrays, plotting, and interacting with shared data catalogues. These tutorials lead into domain-focused “recipes”: self-contained notebooks showing how to perform concrete diagnostics and analyses on ocean and sea-ice datasets. The current repository contains dozens of notebooks grouped into introductory, advanced, and region-specific collections, giving learners an incremental path from first contact with the data ecosystem to adaptation of full workflows for their own science questions.

The Cookbook grew from a practical need inside the COSIMA community: the tools used to analyse modern ocean model output are powerful, but the gap between package-level documentation and a reproducible end-to-end workflow remains large. Users need to understand not only Python and Jupyter (Kluyver et al., 2016), but also how to navigate high-dimensional model output, shared high-performance computing environments, and domain-specific analysis conventions. The Cookbook addresses that gap by packaging reusable workflows in the same

42 medium in which users actually work.

43 Statement of Need

44 Ocean and climate model analysis has a steep entry cost. New users must learn how to find
45 datasets, load them efficiently, interpret metadata, operate on multi-dimensional arrays, and
46 produce scientifically meaningful diagnostics. General-purpose libraries such as xarray (Hoyer &
47 Hamman, 2017) provide the foundations for these tasks, but they do not by themselves show
48 a learner how to translate a research question into a robust analysis workflow for a specific
49 model and computing environment.

50 The COSIMA Cookbook fills that need with a domain-specific, openly maintained collection of
51 computational lessons and examples. Its contribution is not a new analysis package; rather, it
52 is a reusable learning resource that makes existing scientific software, data conventions, and
53 model output more approachable. This aligns well with JOSS's emphasis on open educational
54 resources that enable computational learning through authentic practice.

55 Several design decisions make the Cookbook particularly useful for adoption by other groups.
56 First, each notebook is intended to be self-contained and well-documented, so learners can read,
57 run, and modify a complete workflow without reconstructing missing context from scattered
58 notes. Second, the repository distinguishes tutorials from recipes. Tutorials teach transferable
59 skills, while recipes demonstrate how those skills combine in realistic analysis tasks. Third, the
60 project includes contributor guidance and notebook review conventions that encourage new
61 analyses to be written in a reusable and pedagogically clear style.

62 This structure is valuable beyond the immediate COSIMA context. Many research communities
63 maintain model output on shared infrastructure and face the same challenge: turning expert
64 tacit knowledge into examples that newcomers can adapt. The Cookbook offers a model for
65 how a scientific collaboration can capture that knowledge in version-controlled notebooks,
66 publish it as a living resource, and continuously improve it through community contribution.

67 Educational Design and Experience of Use

68 The learning experience is organised as a progression. The documentation points new users
69 first to introductory lessons, including material on loading, slicing, and visualising model output.
70 Learners then move to more advanced tutorials and finally to recipe notebooks that address
71 concrete scientific questions. The categories of “appetisers”, “mains”, and “local dishes”
72 provide a lightweight pedagogical cue about expected complexity and scope.

73 The intended mode of use is also explicit. The repository is designed around Jupyter-based
74 analysis on the Australian National Computational Infrastructure, with guidance on running
75 notebooks through ARE/JupyterLab sessions and on accessing shared data holdings. This
76 makes the Cookbook more than a static collection of examples: it is operational documentation
77 for a real analysis environment. At the same time, the notebooks demonstrate broadly
78 transferable patterns for working with labelled geophysical data, so individual lessons can be
79 reused or adapted outside that environment.

80 The project also supports community authorship. Contributors are encouraged to submit new
81 notebooks through pull requests, document their methods clearly, and generalise workflows so
82 that they remain useful to others. In that sense, the Cookbook functions both as a learning
83 module for end users and as a framework for teaching reproducible scientific communication
84 through notebook design and peer review.

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